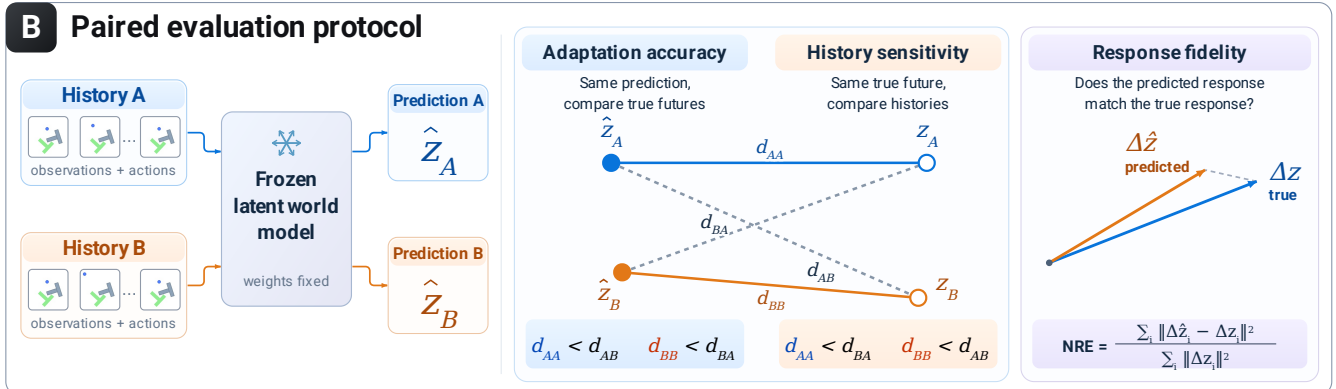

B
Paired evaluation protocol
History A
observations + actions
History B
observations + actions
Frozen latent world model
weights fixed
Prediction A
$\hat{z}_A$
Prediction B
$\hat{z}_B$
Adaptation accuracy
Same prediction, compare true futures
$\hat{z}_A$
z_A
d_{AA}
d_{BA}
d_{AB}
$\hat{z}_B$
z_B
d_{BB}
$d_{AA} < d_{AB}$ $d_{BB} < d_{BA}$
History sensitivity
Same true future, compare histories
$d_{AA} < d_{BA}$ $d_{BB} < d_{AB}$
Response fidelity
Does the predicted response match the true response?
$\Delta\hat{z}$
predicted
Δz
true
NRE = $\frac{\sum_i \|\Delta\hat{z}_i - \Delta z_i\|^2}{\sum_i \|\Delta z_i\|^2}$